

YIRAN PENG

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EDUCATION

University of Edinburgh

Edinburgh, United Kingdom

- **Degree:** MSc Cognitive Science *Sept. 2025 – Present*
- **Core Modules:** Computational Cognitive Science, HCI, Machine Learning Systems, NLP, Computer Graphics

Xi'an Jiaotong-Liverpool University

Suzhou, China

- **Degree:** BEng Computer Science and Technology *Sept. 2021 – Jun. 2025*
- **Core Modules:** Artificial Intelligence, Data Structures, Operating Systems, Computer Networks, Databases, Software Engineering
- **Award:** Second Prize, Asia-Pacific Cup Mathematical Modeling Competition, 2023

PUBLICATIONS

Y. Peng, R. Gong, J. Zhou, T. Chen, Q. Liu, and X. Wang, “Voltage Harmonic Reduction of Cascaded H-Bridge Multilevel Inverter Based on SHEPWM by Using Optimized Immune Algorithm,” in *2023 IEEE 6th Student Conference on Electric Machines and Systems (SCEMS)*, Huzhou, China, 2023, pp. 1–5. doi: [10.1109/SCEMS60579.2023.10379251](https://doi.org/10.1109/SCEMS60579.2023.10379251).

R. Cao, J. He, Y. Peng, Z. Chen, and Q. Sun, “VisionPlay: An Interactive Ball Game Using Computer Vision and Augmented-Reality Overlay,” in *2024 IEEE Smart World Congress (SWC)*, Nadi, Fiji, 2024, pp. 1621–1626. doi: [10.1109/SWC62898.2024.00249](https://doi.org/10.1109/SWC62898.2024.00249).

RESEARCH EXPERIENCE

Multi-Turn Evaluation of Persona Stability and Trust in LLM Agents

Jan. 2026 – Present

Advisor: Jingjie Li

Edinburgh, United Kingdom

- Developed an **HCI framework** to distinguish **acceptable persona evolution** from **harmful drift** in LLM-based agents, and designed a **27-item scale** measuring dynamic trust, expectation violation, and user-initiated repair.
- Built a **scalable multi-turn evaluation pipeline** with **external state tracking**, **controlled growth triggers**, and **trajectory constraints** to quantify changes in agent traits across conversations.
- Compared **three growth mechanisms** across multiple agent roles and designed a **longitudinal study** to assess how persona change affects trust and continued interaction.

GPU Acceleration of GLM-ASR with Triton and FlashAttention

Jan. 2026 – Apr. 2026

Advisors: Luo Mai and Yang Cao

Edinburgh, United Kingdom

- Optimized **GLM-ASR inference** with **custom Triton GPU kernels** and **FlashAttention**, achieving a **1.77× speedup** while maintaining **100% transcription accuracy**.
- Profiled the **end-to-end autoregressive decoding pipeline** and identified **decoder attention** as the primary latency bottleneck.
- Implemented optimized kernels for normalization, activation, RoPE, matrix multiplication, and attention using **online softmax**, **block-wise accumulation**, **kernel fusion**, and **tile tuning**.

AR-Based Remote and Real-Time Manipulation of Unmanned Surface Vehicles

May 2024 – May 2025

Advisors: Xiaohui Zhu and Yong Yue

Suzhou, China

- Implemented a **gaze-based interaction system** on **HoloLens 2**, collecting fixation duration, areas of interest, and gaze latency to support intuitive AR-based vehicle control.
- Applied **gaze-behavior modeling** and **intention inference** to distinguish exploratory from selective attention and mitigate the **Midas Touch problem**.
- Developed a **Unity-based AR visualization system** with **real-time video streaming** for quantitative evaluation of situational awareness and navigation performance.

VisionPlay: Interactive Ball Game with Computer Vision and AR

Feb. 2024 – May 2024

Advisors: Qilei Sun, Quan Zhang, and Jie Sun

Suzhou, China

- Used **Python** and **OpenCV** for **real-time ball tracking and collision detection**, transmitting coordinates to Unity through **socket communication**.
- Developed **Unity** game environments, interaction logic, animations, transitions, and particle effects using **C#**.
- Applied **Intersection over Union** to validate object detection and tracking, achieving over **75% accuracy** across test conditions.
- Conducted **SUS** and **UES evaluations** in **SPSS**; both interactive games scored above **70**, and ANOVA showed significant differences among three game modes ($F(2, 57) = 37.400, p < .001$).

Smart Blue-Algae Treatment System Based on Unmanned Surface Vehicles

Jun. 2024 – Aug. 2024

Advisors: Xiaohui Zhu and Yong Yue

Suzhou, China

- Optimized a **Venturi jet pump** through **CFD simulation** and introduced **three bearings** to reduce friction and improve inhibitor spraying performance.
- Developed a **real-time dosage-control algorithm** using **USV speed, cyanobacteria concentration, river depth, and spray width**.
- Validated the system in a **Dushu Lake field experiment** at a spraying rate of **5 g/s**, reducing cyanobacteria density by approximately **60%**.

Optimization Algorithms for SHEPWM in a Cascaded H-Bridge Inverter

Jun. 2023 – Aug. 2023

Advisors: Xuchen Wang and Qinyao Liu

Suzhou, China

- Implemented an **Immune Algorithm-based SHEPWM method** in **MATLAB** to solve nonlinear equations and determine valid switching angles.
- Designed **affinity evaluation, selection, cloning, mutation, and boundary-condition checks** to improve switching-angle optimization efficiency.
- Compared **Immune Algorithm, PSO, and GA** in **PLECS**; the proposed method reduced total harmonic distortion to approximately **2%** and suppressed targeted line-voltage harmonics.

WORK EXPERIENCE

AI-Assisted Visual Diary with Consistent Character Generation

Jan. 2026 – Present

AI APP Development Intern

Edinburgh, United Kingdom

- Developed an **end-to-end generative visual storytelling pipeline** that transforms diary prompts into personalized ball-jointed-doll images through automated model configuration, generation, and output management.
- Fine-tuned and integrated **custom LoRA models** with **Stable Diffusion** to improve **cross-scene character identity consistency** while reducing visual and stylistic drift.
- Designed **structured LoRA training datasets** and optimized captions, prompt templates, sampling settings, and generation parameters for more controllable character appearance, pose, clothing, and background.

EXTRACURRICULAR ACTIVITIES

Student Accommodation Committee, Xi'an Jiaotong-Liverpool University

Oct. 2021 – Jun. 2023

Vice-Chair

Suzhou, China

- Led the planning and delivery of major student events, including annual club-recruitment activities, coordinating logistics, promotion, and on-site execution.
- Managed the committee's social media channels and produced written and video content to strengthen engagement and improve communication with residents.

SKILLS

Languages: Mandarin Chinese (Native), English (Fluent)

Programming Languages: Python, C++, C#, MATLAB, Java, JavaScript, SQL

Tools and Frameworks: PyTorch, Triton, Stable Diffusion, LoRA, OpenCV, Unity, HoloLens 2, SPSS, Simulink, PLECS, Git